Ans

1) Find out the list of companies who would ask DSA for placement drive.

A) Companies are -

Amazon
Microsoft
google
Adobe
Samsung
Intel
Walmart Global Tech
Oracle
NVIDIA
Pay Pal
American express

2) Find out the gate syllabus for DSA

A DSA syllabus for Gate CSE.

Unit-1: Programming & Data structures:

1) Introduction to DS
2) Arrays
3) linked lists
4) stacks
5) Queues
6) Hash Tables
7) Trees: Binary, BST, AVL, Heap.
8) Graphs
9) Recursion.

Unit 2: Algorithms.

1) Asymptotic Analysis: Time, space complexity, Big-O, Big-Ω, Big-θ Notations.
2) Searching Algorithms: linear & Binary search.
3) Sorting Algorithms: Bubble, selection, Insertion, Merge, Quic Heap sort.
4) Hashing
5) Divide and conquer
6) Greedy Algorithms.
7) Dynamic Programming
8) Graph Algorithms: BFS, DFS

3) Find set of inputs and outputs for following:

i) Electricity bill:

Inputs:
consumer number
consumer name
Previous meter reading
current meter reading
Tariff type
Billing month

Outputs:
Units consumed
energy charges
fixed charges
Taxes
Total bill amount
Due date
Bill status

ii) Bus ticket:

Inputs:
Passenger name
Source
Destination
Travel Date
Number of passengers
Bus type
seat number

Outputs
Ticket number
Bus number
Departure time
Arrival time
Fare
Boarding point

iii) Message to a friend:

Inputs:
Sender name
friend's name
message Text
Date & Time

Outputs:
Message displayed
Sent status
delivery Time
Read status

4) Today when you started from home to college, what are the various data structures you identified in your surroundings

A) Data structures

	Real-life example
Array	Row of parked bikes and cars
Queue	line of people at bus stop
Stack	Stack of books
Graph	Road connecting different places
linked list	Bus route map
Hash table	Dictionary/Phone contacts
Tree	Branches of tree or college hierarchy
Matrix (2D array)	classroom seating arrangement
Priority Queue	Ambulance in traffic
Circular Queue	Vehicles moving around a traffic roundabout.